SRS/SDD Self-Review Report – Review 2

Form BM02. Run the standard prompt on the proposed final registration form and the repository at tag review-2. Keep every finding exactly as produced by the AI; add only the Team response column. Export to PDF. File name: SWP490_<ProjectCode>_R2_SelfReview.pdf

1. Run Information

Project code	GFA26SE117 / FA26SE069
Project name	Development of ECHO PROTOCOL: An Online Cooperative Horror Game with Adaptive AI and a Client-Server Virtual Item Shop
Team	Đặng Huy Hoàng; Hứa Thành Đạt; Lâm Thanh Phúc; Nguyễn Nhật Lệ Thuyên
Supervisor	Đào Quốc Tuấn
AI tool and model	Codex Agent : model GPT-6 Sol
Run date	31/10/2026
Repository tag / commit	Pre-tag review: LTP @ 08de8d2347d118077a076ba032568236efac4898
Registration form version used	FA26SE069_DEVELOPMENT_OF_ECHO_PROTOCOL_AN_ONLINE_C_tuandq19
Version	1.0
Date	21/10/2026

2. Standard Prompt

Use this prompt without modification.

```
You are reviewing the SRS and SDD of a software capstone project against its
registration form and source code. Report only; do not change any file.

Inputs: the registration form, SRS, SDD (with diagrams), source code and
database script in this repository.

Check:
1. Scope: FRs in the SRS not covered by the registration form, and
   registration items with no FR.
2. Requirements: FRs without an actor or testable acceptance criteria;
   NFRs without metric, threshold and measurement method.
3. Data: entities missing from the Logical ERD; ERD entities or attributes
   missing from the database script, and tables not in the ERD.
4. States: for each main entity, states used in code but missing from its
   state machine; transitions with no triggering FR; states with no exit.
5. Flows: main business flows in activity diagrams with no implementation
   or no test.
6. Architecture: components or third-party services in code but not in the
   architecture diagram, and the reverse.
7. Design vs code: modules, APIs or business rules implemented differently
   from the SDD.

Output one table: ID | Category | Location | Finding | Severity
(High/Medium/Low) | Evidence. Then list the 5 most important findings.
If something cannot be verified from the repository, say so; do not guess.
```

3. Findings and Team Response

Team response: Fixed (link to commit) · Kept (reason) · Planned (owner and due date). Do not delete any finding.

ID	Category	Location	Finding	Severity	Evidence	Team response
F01	Scope / design vs code	Registration §3.2(c); SRS FR-AI-04; SDD Figure 25	The registered UV counterplay interrupts the Stalker under valid conditions, but the SRS and SDD say UV does not interrupt the Expedition Stalker. The documented acceptance target therefore conflicts with the registration.	High	[Registration](E:/ECHO-PROTOCOL/docs/review-2/Registration/Submitted_Registration_2026-10-21.docx), [SRS](E:/ECHO-PROTOCOL/docs/review-2/SRS/Report3_SRS.docx), [WF-03](E:/ECHO-PROTOCOL/docs/review-2/diagrams/workflow/WF-03_Stalker_AI_Gameplay_Workflow.mmd)	Kept – Documented Gameplay Design Deviation The submitted registration form is retained unchanged as the approved project baseline. During gameplay refinement, the team revised the UV interaction with the Expedition Stalker. UV equipment remains available under configured resource and activation constraints, but it does not stun, repel, or interrupt the Stalker. This design decision preserves the Stalker as a persistent survival-horror threat and emphasizes stealth, movement, environmental awareness, and cooperative survival. The difference from the registered UV-interruption behaviour is explicitly acknowledged as a gameplay design deviation, not treated as a requirement already satisfied. The final behaviour is consistently specified in SRS FR-AI-04 and the SDD Stalker gameplay workflow. The team will present this deviation for academic review and acceptance. Resolution Evidence: SRS FR-AI-04; SDD WF-03 – Stalker AI Gameplay Workflow; submitted Registration §3.2(c).
F02	Scope	SRS FR-AUT-05; registration account requirements	Mandatory versioned legal acceptance and its gameplay/shop access gate have no explicit item in the submitted registration.	Medium	[SRS](E:/ECHO-PROTOCOL/docs/review-2/SRS/Report3_SRS.docx), [Registration](E:/ECHO-PROTOCOL/docs/review-2/Registration/Submitted_Registration_2026-10-21.docx)	Kept – Additional Account and Access Control - The submitted registration form defines account authentication, authorized access, and backend protection of critical game operations. The versioned Content Warning and Legal Agreements acknowledgement was introduced as an additional access-control requirement within this existing account and authentication scope. - The implementation does not introduce a separate gameplay mode or commercial payment requirement. It provides a combined first-use acknowledgement screen, stores acceptance against the authenticated account and current policy version, and restricts protected operations when acceptance is missing or outdated. - The team retains this additional control as a documented refinement of the registered account and security requirements. Its implementation and acceptance criteria are specified in SRS FR-AUT-05 and supported by the backend agreement endpoints and middleware. - Resolution Evidence: SRS FR-AUT-05; SDD

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						authentication and legal-acceptance flow; AgreementsController.cs; AgreementGateMiddleware.cs.
F03	Scope	Registration §3.2(f); SRS FR-ECO-06	The registration lists administration of item ownership; the SRS admin FR specifies viewing user, wallet and purchase records and adjusting wallets, with no admin ownership action or acceptance criterion.	Medium	[Registration](E:/ECHO-PROTOCOL/docs/review-2/Registration/Submitted_Registration_2026-10-21.docx), [SRS](E:/ECHO-PROTOCOL/docs/review-2/SRS/Report3_SRS.docx)	Fixed – Requirement Responsibility and Traceability Clarified - The team clarified the distinction between backend item ownership management and Limited Administration responsibilities. - Item ownership is managed by the Shop and Inventory backend through validated purchase transactions, covered by SRS FR-ECO-02 and FR-ECO-03. Limited Administration, covered by FR-ECO-06, supports authorized user, wallet, and purchase-record access and auditable wallet adjustments. - Direct administrative ownership modification is not part of the mandatory implementation. The SRS was updated to explicitly document this responsibility boundary and its relationship to the submitted registration. - Resolution Evidence: SRS §3.8, FR-ECO-02, FR-ECO-03, and FR-ECO-06; Registration §3.2(f); backend AdminController.cs and Shop/Inventory services.
F04	Requirements	SRS FR-EXP-01, FR-ADA-05, FR-AI-05, FR-UI-05	FR-EXP-01 and FR-ADA-05 assign work to the deliverable/team rather than an identified system actor. FR-AI-05's "no blocker" and FR-UI-05's "clear" feedback lack defined pass/fail conditions.	Medium	[SRS §3](E:/ECHO-PROTOCOL/docs/review-2/SRS/Report3_SRS.docx)	F04 — Team Response: Fixed Updated FR-EXP-01, FR-ADA-05, FR-AI-05, and FR-UI-05 to clarify responsible actors or evaluation roles and establish explicit, testable acceptance and negative-case conditions. The original requirement IDs and registration traceability were preserved. Resolution Evidence: docs/review-2/SRS/Report3_SRS.docx, Sections 3.5, 3.6, 3.7, and 3.9.
F05	Requirements	SRS §§4.2.1–4.2.2, NFR-05	Usability and reliability statements have no quantified metric, threshold and measurement method. NFR-05 defines a restore rehearsal but explicitly leaves RPO/RTO unspecified, so recovery performance cannot be accepted against a threshold.	Medium	[SRS §4](E:/ECHO-PROTOCOL/docs/review-2/SRS/Report3_SRS.docx)	F05 — Team Response: Fixed Revised the Usability and Reliability quality attributes to include measurable acceptance thresholds and verification methods. Updated NFR-05 to define recovery integrity acceptance based on predefined validation samples while explicitly excluding unsupported production RPO/RTO guarantees. Resolution Evidence: docs/review-2/SRS/Report3_SRS.docx, Sections 4.2.1, 4.2.2, and NFR-05.

ID	Category	Location	Finding	Severity	Evidence	Team response
F06	Data	SDD Figure 5 logical ERD; migration script	The ERD names ScenarioConfigDefinitions, but the script creates ScenarioConfigs; the ERD's named entity is absent from the database script.	High	[Figure 5 SQL:187 (line 187)](E:/ECHO-PROTOCOL/docs/review-2/diagrams/erd_reference/ECHO_PROTOCOL_Figure_05_LTP_ERD_REVIEWED.sql:187), [migration script:819 (line 819)](E:/ECHO-PROTOCOL/docs/review-2/database/postgresql_migrations.sql:819)	F06 — Fixed - Updated Figure 5 to align the scenario configuration entity with the actual PostgreSQL migration schema. Renamed ScenarioConfigDefinitions to ScenarioConfigs, corrected the composite foreign-key reference from ScenarioDecisions, and added the missing configuration attributes, including DetectionFillRate, DetectionDecayRate, RouteModifier, and Provenance. - The database migration itself was not modified. The updated ERD was regenerated and incorporated into the SDD. - Resolution Evidence: Figure 5 ERD SQL; SDD Figure 5; PostgreSQL migration AddScenarioConfigRegistryM4043M4045.
F07	Data	SDD Figures 4–5 logical ERDs; migration script	MatchScores and ScenarioContentDefinitions appear in the script and SDD table descriptions but have no entity in either selected logical ERD. The script also has payment and AED ledger tables absent from those ERDs. The ERD SQL calls itself a selected view, so this is a documentation coverage gap, not evidence that the tables are unintended.	Medium	[SDD §2.2](E:/ECHO-PROTOCOL/docs/review-2/SDD/Report4_SDD.docx), [migration script:674 (line 674)](E:/ECHO-PROTOCOL/docs/review-2/database/postgresql_migrations.sql:674), [migration script:858 (line 858)](E:/ECHO-PROTOCOL/docs/review-2/database/postgresql_migrations.sql:858), [Figure 5 SQL](E:/ECHO-PROTOCOL/docs/review-2/diagrams/erd_reference/ECHO_PROTOCOL_Figure_05_LTP_ERD_REVIEWED.sql)	F07 — Fixed - Updated the selected logical ERD to include MatchScores and ScenarioContentDefinitions, which were previously described in the SDD but absent from the corresponding diagram. - Added an entity coverage and scope mapping to SDD Section 2.2.2 to distinguish mandatory Review 2 persistence entities from payment integrations and additional Adaptive AI research or pilot ledgers. - The documentation now explicitly identifies the selected ERD boundaries and refers to the PostgreSQL migration script as the authoritative source for physical table definitions and relationships. - Resolution Evidence: Updated Figure 5; SDD Sections 2.2 and 2.2.2; PostgreSQL migration script.
F08	States	SDD Figure 13; Unity match code	Puzzle and SecurityHold are used in match interaction code but have no transitions in the mission state machine. Its note acknowledges the enum values while	Medium	[Figure 13](E:/ECHO-PROTOCOL/docs/review-2/diagrams/SDD/Figure_13_Expedition_Mission_State_Machine.mmd),	F08 — Fixed - Revised Figure 13 to show the active Host-authoritative Expedition phase progression and the nested Zone2MissionStage transitions. The diagram now includes terminal discovery, relay repair, Security Hold readiness and execution, authorization-code generation, and Zone 2

ID	Category	Location	Finding	Severity	Evidence	Team response
			leaving their entry, exit and triggering FRs undefined.		[NetworkSectorBox.cs (line 214)](E:/ECHO-PROTOCOL/KLTN/Assets/_Project/Scripts/Networking/Interaction/NetworkSectorBox.cs:214), [NetworkMatchState.cs (line 62)](E:/ECHO-PROTOCOL/KLTN/Assets/_Project/Scripts/Networking/Match/NetworkMatchState.cs:62)	completion. - Clarified that Puzzle and SecurityHold remain legacy NetworkMatchPhase enumeration values and are not standalone phases of the selected mandatory Expedition route. The SDD description was updated to distinguish active mission transitions from legacy compatibility definitions. - Resolution Evidence: Updated Figure 13; SDD Section 3.4; NetworkMatchState.cs; Zone2MissionStage.cs.
F09	States	SDD Figure 11; SRS FR-PLY-03/FR-AI-03	The player state machine includes Alive → Caught → Downed/Eliminated, but the cited mandatory FRs specify Downed/elimination and attacks without defining a catch trigger or acceptance case. The SDD labels catch behavior mode dependent. No unintended nonterminal state with no exit was verified in the three supplied player, mission and Stalker state machines.	Low	[Figure 11](E:/ECHO-PROTOCOL/docs/review-2/diagrams/SDD/Figure_11_Player_Life_State_Machine.mmd), [SRS §3](E:/ECHO-PROTOCOL/docs/review-2/SRS/Report3_SRS.docx), [NetworkPlayerLifeState.cs (line 9)](E:/ECHO-PROTOCOL/KLTN/Assets/_Project/Scripts/Networking/Player/NetworkPlayerLifeState.cs:9)	F09 — Fixed Clarified the distinction between mandatory Expedition player life-state transitions and shared, mode-dependent capture behaviour. Figure 11 and SDD Section 3.3 were updated to identify Caught as an optional shared state associated with validated ghost capture behaviour. The WF-03 State Changes description was also corrected to remove Caught from the mandatory Stalker encounter progression. No additional Stalker capture requirement or gameplay implementation was introduced. Resolution Evidence: Updated Figure 11; SDD Sections 3.3 and 3.10.3; NetworkPlayerLifeState.cs.
F10	Flows	SDD §3.10, WF-01–WF-04	The SDD names TC-WF01–TC-WF04 tests, but those identifiers and a Report 5 execution record were not found in the repository. Component and backend tests exist; their presence does not verify the four complete workflows, especially a full Expedition Win/Lose run. End-to-end execution cannot be verified from this repository.	High	[SDD §3.10](E:/ECHO-PROTOCOL/docs/review-2/SDD/Report4_SDD.docx), [WF-02](E:/ECHO-PROTOCOL/docs/review-2/diagrams/workflow/WF-02_Expedition_Objectives_Outcome.mmd), [existing match-result report](E:/ECHO-PROTOCOL/docs/M4_009_TEST_REPORT.md)	Fixed – Workflow Test Evidence Completed The team completed and documented end-to-end verification for WF-01 through WF-04, covering authentication and lobby operations, Expedition Win/Lose conditions, Stalker AI behaviour and player survival, and match result/reward/economy workflows. The executed test cases are supported by recorded testing videos and relevant execution records. Evidence links, test environments, source commit information, observed results, and execution statuses are documented in the workflow test traceability record. The available evidence was reviewed against the corresponding SDD workflows and acceptance criteria. Any remaining test failures or limitations are documented separately rather

ID	Category	Location	Finding	Severity	Evidence	Team response
						than reported as successful execution. Resolution Evidence: WF-01–WF-04 testing videos and docs/review-2/evidence/WF_TEST_TRACEABILITY.md.
F11	Architecture	SDD Figure 1; backend and web source	The architecture diagram omits implemented payOS checkout and the player web portal. Both are explicitly outside registered mandatory scope, but they are present as API services, payment pages and a provider integration in the reviewed source.	Medium	[Figure 1](E:/ECHO-PROTOCOL/docs/review-2/diagrams/SDD/Figure_01_Overall_System_Architecture - Sao chép (2\).mmd), [Program.cs (line 249)](E:/ECHO-PROTOCOL/EchoProtocol.Backend/src/EchoProtocol.Api/Program.cs:249), [wallet page](E:/ECHO-PROTOCOL/EchoProtocol.Web/src/app/(player\)/wallet/page.tsx), [Registration](E:/ECHO-PROTOCOL/docs/review-2/Registration/Submitted_Registration_2026-10-21.docx)	F11 — Fixed Updated Figure 1 to document the implemented Player Web Portal and payOS integration as additional architecture components outside mandatory Review 2 acceptance. The SDD architecture description now distinguishes the required Unity, Photon Fusion, backend and persistence components from additional player web and payment functionality. The update documents existing source integrations without expanding mandatory acceptance scope or claiming production payment readiness. Resolution Evidence: Updated Figure 1; SDD Sections 1.1 and 1.1.4; PaymentOrdersController.cs; PaymentWebhooksController.cs; Player Web Portal source.
F12	Design vs code	WF-04 reward lookup	WF-04 labels the reward request GET /reward/me; the implemented route requires a match ID: GET /api/matches/{matchId}/reward/me. A test derived literally from the diagram would call the wrong endpoint.	Low	[WF-04:10 (line 10)](E:/ECHO-PROTOCOL/docs/review-2/diagrams/workflow/WF-04_Match_Reward_Economy_Inventory.mmd:10), [MatchResultsController.cs (line 93)](E:/ECHO-PROTOCOL/EchoProtocol.Backend/src/EchoProtocol.Api/Controllers/MatchResultsController.cs:93), [MatchAuthorityApiService.cs (line 103)](E:/ECHO-PROTOCOL/KLTN/	F12 — Fixed Corrected the WF-04 reward lookup endpoint from GET /reward/me to GET /api/matches/{matchId}/reward/me. The workflow diagram and the embedded SDD Figure 26 were updated to match the implemented MatchResultsController route. Resolution Evidence: Updated WF-04 Mermaid; SDD Figure 26; MatchResultsController.cs.

ID	Category	Location	Finding	Severity	Evidence	Team response
					Assets/_Project/Scripts/Networking/Authority/MatchAuthorityApiService.cs:103)	

4. Top 5 Findings (as reported by the AI)

#	Finding ID	Summary
1	F01	The submitted registration requires conditional UV interruption, while the mandatory Expedition Stalker design explicitly excludes UV stun or interruption.
2	F10	Complete end-to-end test execution evidence for WF-01 through WF-04 could not be verified from the repository.
3	F06	The logical ERD uses ScenarioConfigDefinitions, while the PostgreSQL migration defines ScenarioConfigs, creating a database design mismatch.
4	F07	The selected logical ERDs omit implemented database entities, including MatchScores, ScenarioContentDefinitions, and additional payment and Adaptive AI ledger tables.
5	F05	Usability and reliability requirements lack measurable acceptance criteria, while recovery requirements do not define formal RPO/RTO thresholds.

5. Summary

Severity	Total	Fixed	Kept	Planned
High	3	1	1	1
Medium	7	6	1	0
Low	2	2	0	0

Every High finding marked Kept or Planned must also appear in section 5 (Known Gaps) of the Design Brief.